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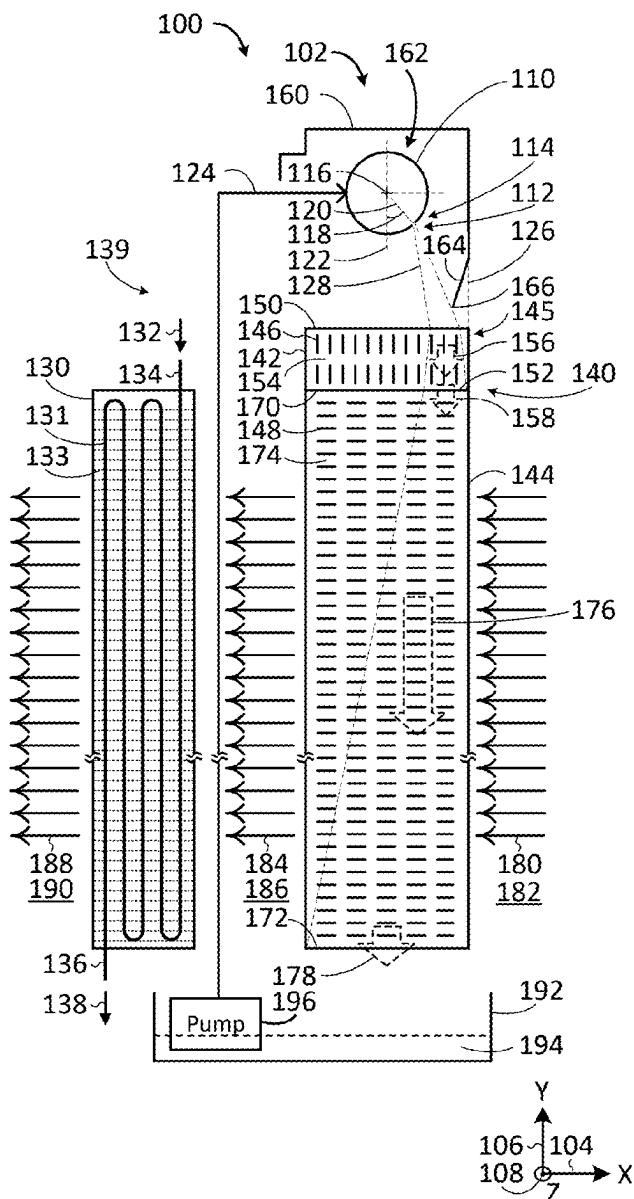
(19) **United States**(12) **Patent Application Publication**
Zha(10) **Pub. No.: US 2025/0277592 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **ADIABATIC PRE-COOLING FOR A GAS COOLER**(52) **U.S. Cl.**CPC **F24F 5/0035** (2013.01)(71) Applicant: **Heatcraft Refrigeration Products LLC**, Stone Mountain, GA (US)(72) Inventor: **Shitong Zha**, Snellville, GA (US)(21) Appl. No.: **18/593,264**(22) Filed: **Mar. 1, 2024****Publication Classification**(51) **Int. Cl.**
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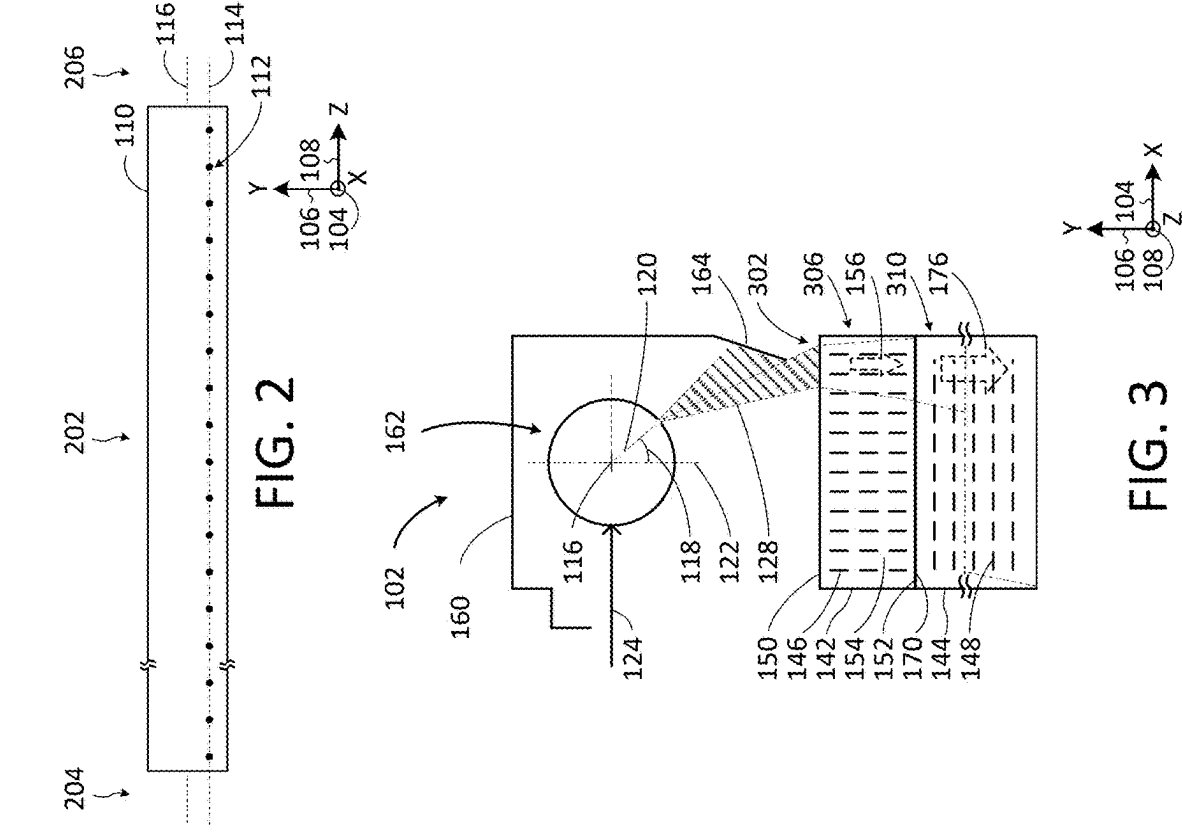
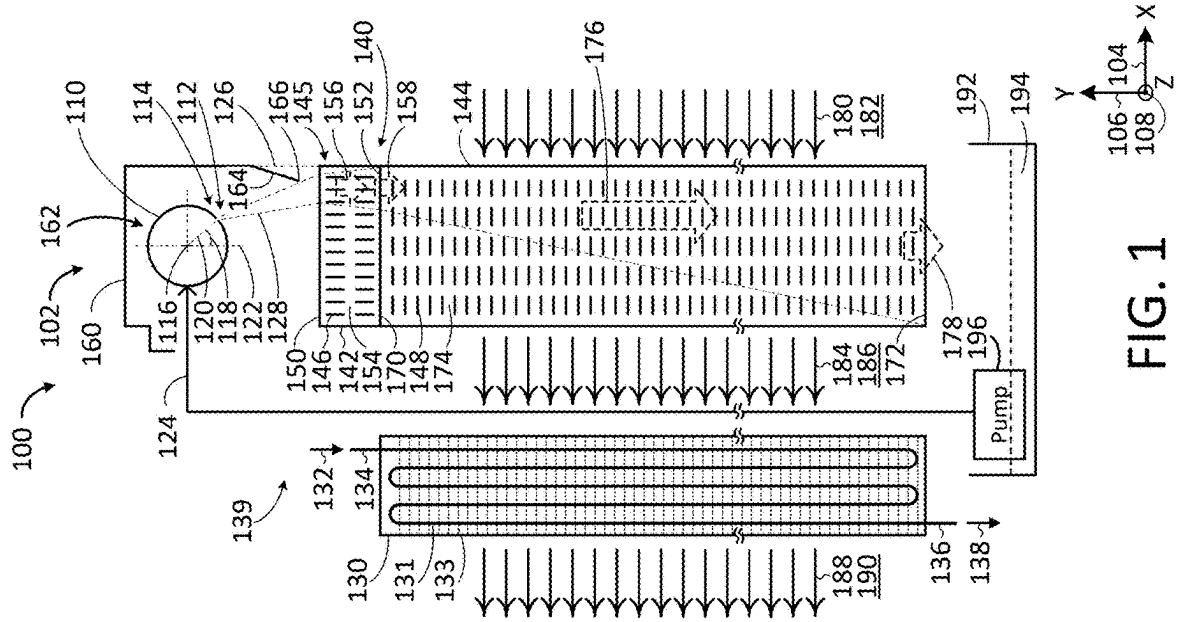
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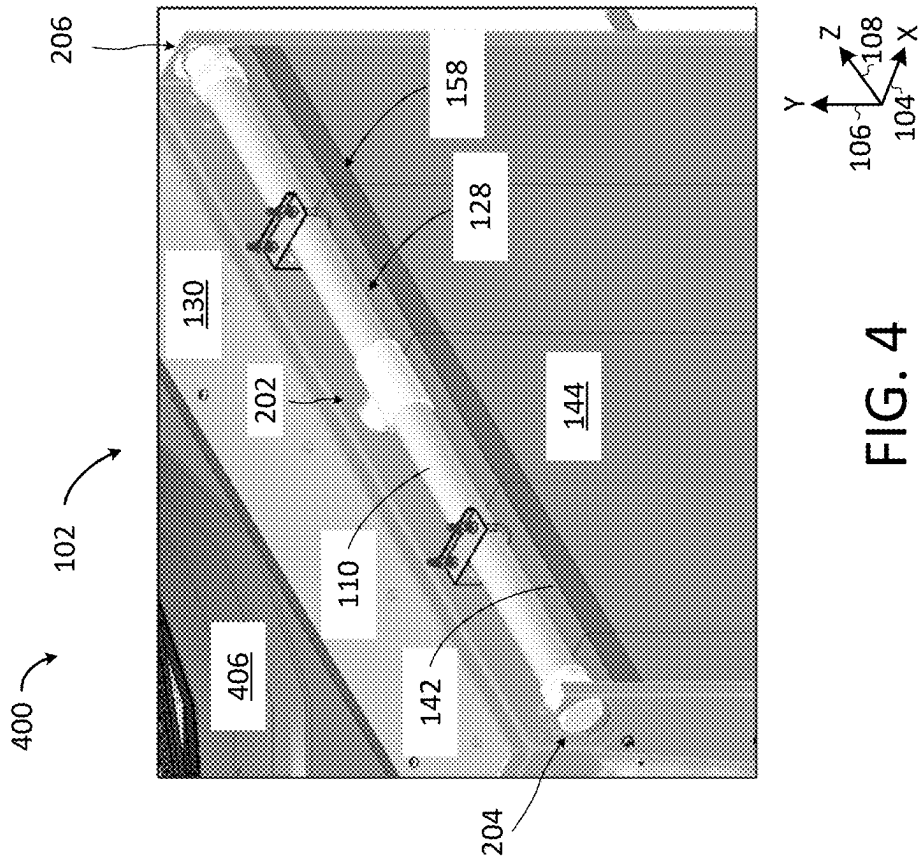
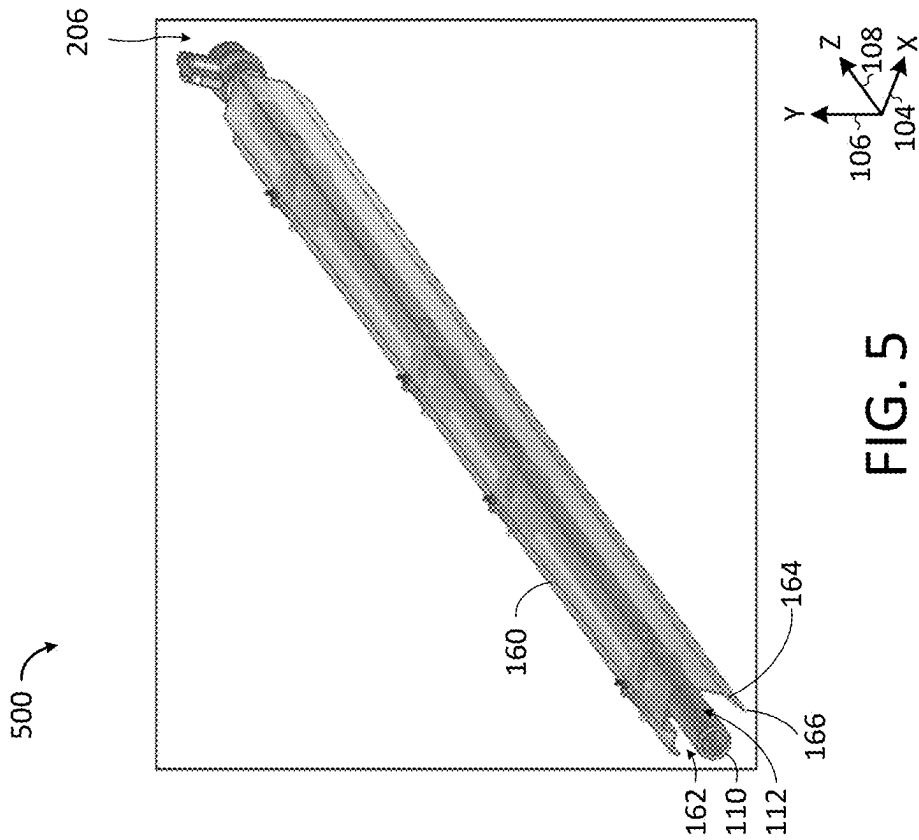
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ABSTRACT

A cooling device may include a water header pipe with drain holes arranged in a row oriented at an angle. The water header pipe may receive a supply of water and discharge a first portion of water through the drain holes. An evaporative cooling medium may include a first media portion and a second media portion. The first media portion may have a first media body to receive the first portion of water and discharge a third portion of water. The second media portion may have a second media body to receive the third portion of water and conduct a fourth portion of water. The second media body may receive a first inflow of air at a first temperature and discharge a first outflow of air at a second temperature that is lower than the first temperature by evaporative cooling of the fourth portion of water.







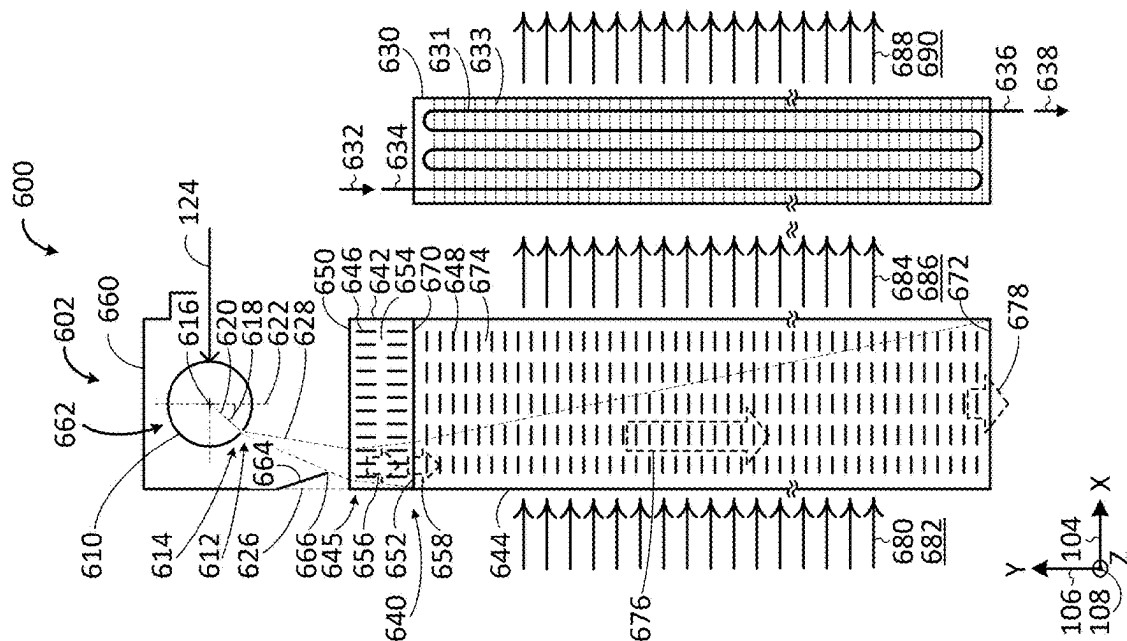


FIG. 6

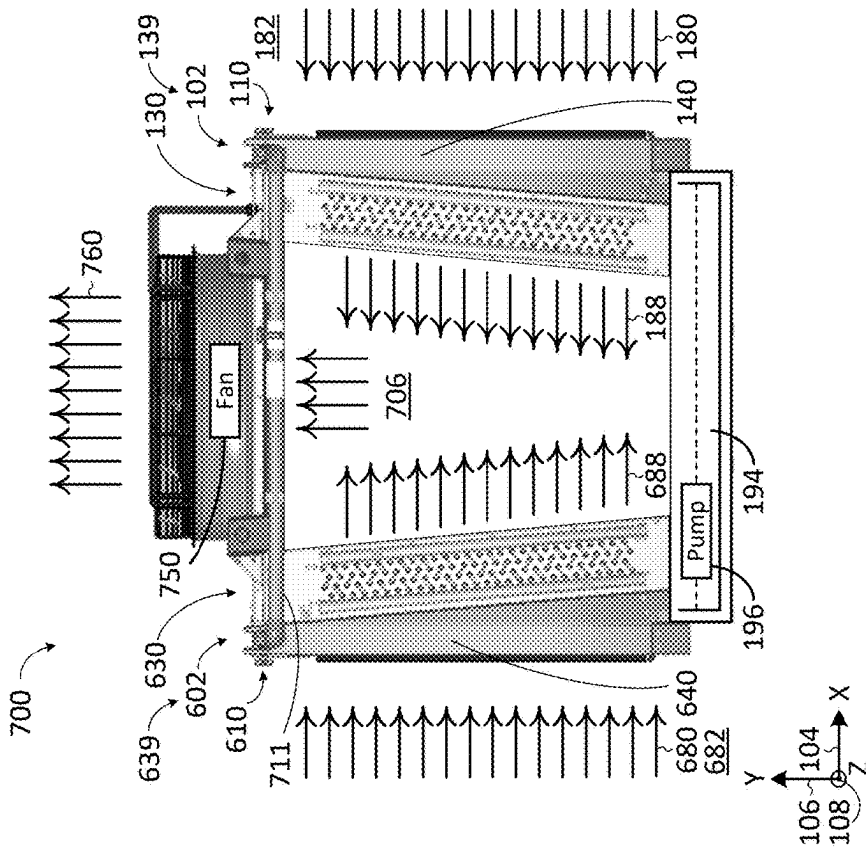
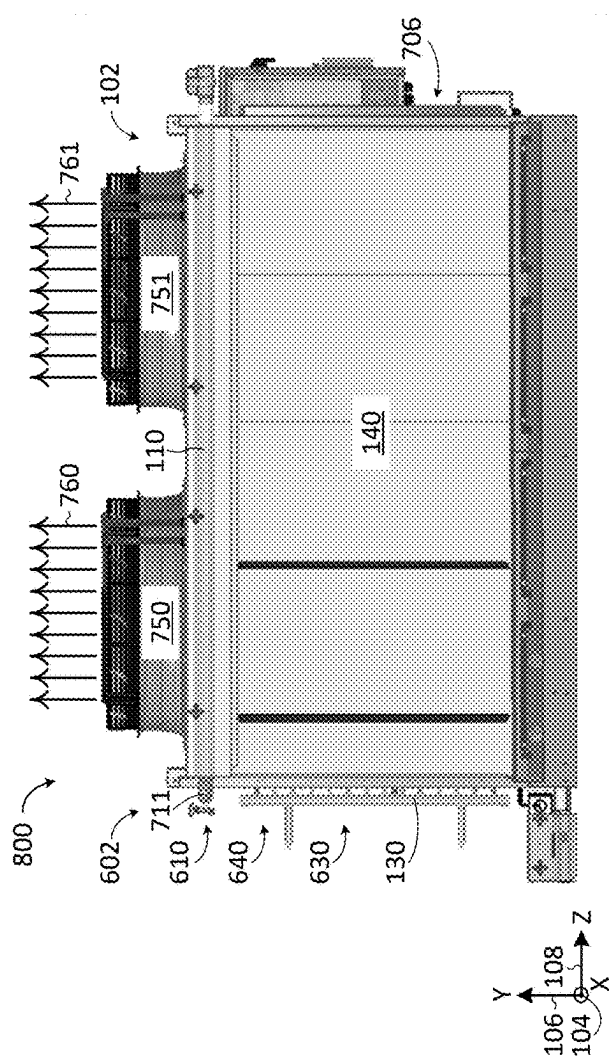


FIG. 7



8
G.
F.

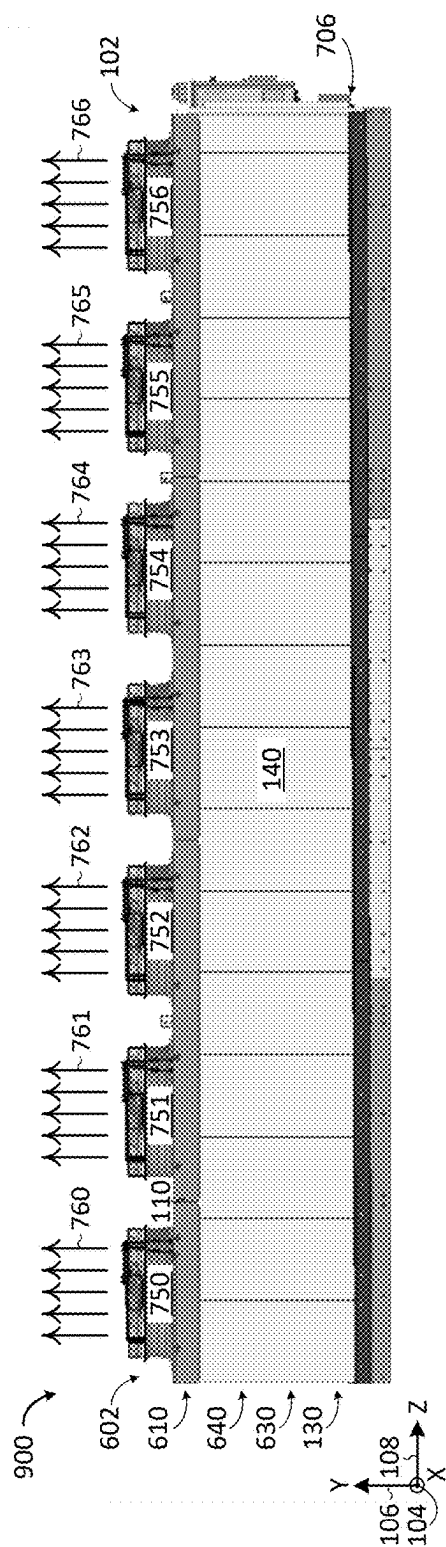


FIG. 9

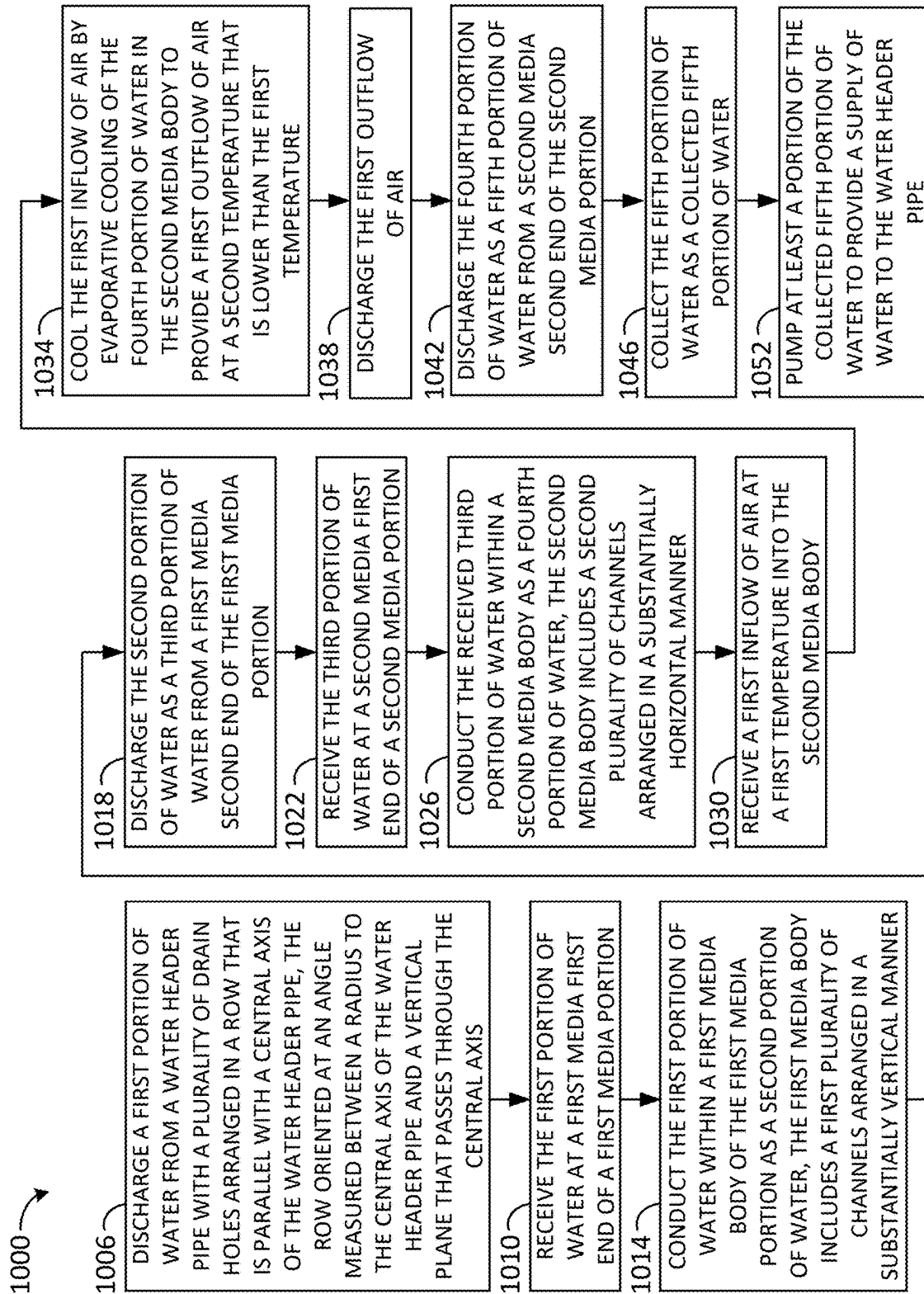


FIG. 10

ADIABATIC PRE-COOLING FOR A GAS COOLER

TECHNICAL FIELD

[0001] This disclosure relates generally to cooling systems and more particularly to improved adiabatic pre-cooling for a gas cooler.

BACKGROUND

[0002] Gas cooling systems are used to reduce the temperature of a circulating coolant in gas form, typically without changing the state of the coolant from gas to liquid. Circulating coolant in hot gas form, such as carbon dioxide or other materials and compounds, may be heated in various ways such as by a compressor, by passing through a heat exchanger, by absorbing heat from warmer region such as an interior space of a home or office building, by other heat generating or heat transferring processes and systems, or a combination of these. A hot gas may pass through a pipe with attached fins so that heat from the hot gas may conduct to the attached fins and be transferred to a lower temperature environment by convection and/or radiation so that the temperature of the hot gas is reduced.

[0003] Adiabatic cooling has been used to provide additional cooling in various applications, where a water header pipe may discharge water onto a portion of an evaporative cooling pad by discharging the water in a directly downward direction in order to maximize drainage from the water header pipe, for example.

SUMMARY

[0004] To improve gas cooling, this disclosure provides various cooling systems, cooling devices, and cooling methods that utilize adiabatic cooling.

[0005] According to an example, a cooling device may include a water header pipe having a plurality of drain holes arranged in a row that may be parallel with a central axis of the water header pipe, the row may be oriented at an angle measured between a radius to the central axis of the water header pipe and a vertical plane passing through the central axis, the water header pipe may be configured to receive a supply of water and discharge a first portion of water through the plurality of drain holes; and an evaporative cooling medium may comprise a first media portion and a second media portion, the first media portion may have a first media first end, a first media second end, and a first media body, the first media first end may be configured to receive the first portion of water, the first media body may be configured to conduct the received first portion of water within the first media body as a second portion of water, the first media second end may be configured to discharge a third portion of water; and the second media portion may have a second media first end, a second media second end, and a second media body, the second media first end may be configured to receive the third portion of water, the second media body may be configured to conduct the received third portion of water within the second media body as a fourth portion of water, wherein the second media body may be configured to receive a first inflow of air at a first temperature and discharge a first outflow of air at a second temperature that is lower than the first temperature, the first outflow of air may be cooled by evaporative cooling of the fourth portion of water.

[0006] According to this example, the cooling device wherein the angle may be between at least 10-degrees to not more than 40-degrees. The cooling device wherein the first portion of water may be directed to a first media portion exterior edge, and wherein the second portion of water may be configured to initially saturate a second media portion exterior surface. The cooling device wherein the first media body may include a first plurality of channels arranged in a substantially vertical manner, and wherein the second media body may include a second plurality of channels arranged in a substantially horizontal manner, orthogonal to the first plurality of channels. The cooling device wherein at least one of the first media portion and the second media portion may include cellulose. The cooling device may further include an elongated deflector shield disposed to at least partially surround an upper portion of the water header pipe, the elongated deflector shield may be configured to at least one of protect the upper portion of the water header pipe and redirect water from the water header pipe in a direction toward the first media first end.

[0007] According to this example, the cooling device wherein the elongated deflector shield may include a lower portion with a lower edge portion that may be laterally displaced from a second vertical plane that may be coplanar with an outer surface of the first media portion. The cooling device wherein the second media second end may be configured to discharge a fifth portion of water, the device may further include a reservoir configured to collect the fifth portion of water as a collected fifth portion of water; and a pump configured to transport the collected fifth portion of water and provide the supply of water to the water header pipe. The cooling device wherein the pump may have a pump profile that is limited to at least one of avoid overspray, avoid bouncing off (e.g., ricochet) first media first end, and promote capture and conduction of water in the first media portion. The cooling device wherein the supply of water may be provided to one of a center portion of the water header pipe and an end portion of the water header pipe, wherein at least one end portion of the water header pipe may be closed.

[0008] According to this example, a cooling system may include the cooling device and may further include a gas cooler disposed adjacent to the evaporative cooling medium and configured to receive the first outflow of air and provide a second outflow of air. According to this example, a cooling system may include the cooling device, wherein the water header pipe may a first water header pipe and the evaporative cooling medium may be a first evaporative cooling medium. According to this example, the cooling system may further include a second water header pipe having a second plurality of drain holes arranged in a second row that is parallel with a second central axis of the second water header pipe, the second row may be oriented at a second angle measured between a second radius to the second central axis of the second water header pipe and a second vertical plane passing through the second central axis, the second water header pipe may be configured to receive the supply of water and discharge a sixth portion of water through the second plurality of drain holes.

[0009] According to this example, this system may include a second evaporative cooling medium that may include a third media portion and a fourth media portion, the third media portion may have a third media first end, a third media second end, and a third media body, the third media

first end may be configured to receive the sixth portion of water, the third media body may be configured to conduct the received sixth portion of water within the third media body as a seventh portion of water, the third media second end may be configured to discharge an eighth portion of water; and the fourth media portion may have a fourth media first end, a fourth media second end, and a fourth media body, the fourth media first end may be configured to receive the eighth portion of water, the fourth media body may be configured to conduct the received eighth portion of water within the fourth media body as a ninth portion of water, wherein the fourth media body may be configured to receive a second inflow of air at a fourth temperature and discharge a third outflow of air at a fifth temperature that may be lower than the fourth temperature, the third outflow of air may be cooled by evaporative cooling of the ninth portion of water.

[0010] According to this example, the cooling system wherein the second angle may be between at least 10-degrees to not more than 40-degrees, wherein the sixth portion of water may be directed to a third media portion exterior edge, wherein the eighth portion of water may be configured to initially saturate a fourth media portion exterior surface, wherein at least one of the third media portion and the fourth media portion may include cellulose. The cooling system may further include a second elongated deflector shield disposed to at least partially surround a second upper portion of the second water header pipe, the second elongated deflector shield may be configured to at least one of protect the second upper portion of the second water header pipe and redirect water from the second water header pipe in a direction toward the third media first end, wherein the second elongated deflector shield may include a second lower portion with a second lower edge portion that may be laterally displaced from a fourth vertical plane that is coplanar with a second outer surface of the third media portion. The cooling system may further include a first gas cooler disposed adjacent to the first evaporative cooling medium and configured to receive the first outflow of air and provide a second outflow of air, a second gas cooler disposed adjacent to the second evaporative cooling medium and configured to receive the third outflow of air and provide a fourth outflow of air. The cooling system may further include at least one fan unit configured to receive the second outflow of air and the fourth outflow of air and expel a fifth outflow of air.

[0011] According to an example, a cooling method may include discharging a first portion of water from a water header pipe with a plurality of drain holes arranged in a row that is parallel with a central axis of the water header pipe, the row oriented at an angle measured between a radius to the central axis of the water header pipe and a vertical plane that passes through the central axis. The cooling method may include receiving the first portion of water at a first media first end of a first media portion. The cooling method may include conducting the first portion of water within a first media body of the first media portion as a second portion of water, the first media body may include a first plurality of channels arranged in a substantially vertical manner. The cooling method may include discharging the second portion of water as a third portion of water from a first media second end of the first media portion. The cooling method may include receiving the third portion of water at a second media first end of a second media portion. The cooling method may include conducting the third portion of

water within a second media body as a fourth portion of water, the second media body may include a second plurality of channels arranged in a substantially horizontal manner. The cooling method may include receiving a first inflow of air at a first temperature into the second media body. The cooling method may include cooling the first inflow of air by evaporative cooling of the fourth portion of water in the second media body to provide a first outflow of air at a second temperature that is lower than the first temperature. The cooling method may include discharging the first outflow of air.

[0012] According to this example, the cooling method wherein the angle may be between at least 10-degrees to not more than 40-degrees. The cooling method wherein the first portion of water may be directed to a first media portion exterior edge, and wherein the fourth portion of water may be configured to initially saturate a second media portion exterior surface. The cooling method may further include discharging the fourth portion of water as a fifth portion of water from a second media second end of the second media portion. The cooling method may include collecting the fifth portion of water as a collected fifth portion of water. The cooling method may include pumping at least a portion of the collected fifth portion of water to provide a supply of water to the water header pipe.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] For a more complete understanding of the present disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawing figures, in which:

[0014] FIG. 1 illustrates a side plan view of a cooling system incorporating a first cooling device, according to an example;

[0015] FIG. 2 illustrates a front plan view of a water header pipe, according to an example;

[0016] FIG. 3 illustrates an enlarged side plan view of a portion of the cooling system of FIG. 1, according to an example;

[0017] FIG. 4 illustrates a partial perspective view of a cooling system incorporating the cooling device of FIG. 1, according to an example;

[0018] FIG. 5 illustrates a partial perspective view of an elongated deflector shield partially surrounding an upper portion of the water pipe of the cooling system of FIG. 1, according to an example;

[0019] FIG. 6 illustrates a side plan view of a cooling system incorporating a second cooling device, according to an example;

[0020] FIG. 7 illustrates an end view of a cooling system incorporating two cooling devices, according to an example;

[0021] FIG. 8 illustrates a side view of a dual-fan cooling system incorporating two cooling devices, according to an example;

[0022] FIG. 9 illustrates a side view of a 7-fan cooling system incorporating two cooling devices, according to an example; and

[0023] FIG. 10 illustrates a method of operating a cooling system, according to an example.

DETAILED DESCRIPTION

[0024] As will be described more fully below, an improved adiabatic cooler may include a water header pipe

having a discharge point of water from a plurality of drain holes at angle that may not be directly downward. Instead, the discharge point may be disposed at an angle and may be directed at least partly towards an outside edge of an adjacent cooling pad, for example. Various advantages from this arrangement may include reducing the likelihood that dirt or other contaminants that may reside or collect in a supply of water or which may accumulate in a bottom portion of a water header pipe may block passage of water through the plurality of drain holes. Further, displacing the plurality of drain holes from a directly downward row position may provide an exterior surface of the water header pipe that may be easier to clean, be less likely to corrode, and be more long-lasting.

[0025] Hereinafter, a description will be given of respective examples of the present disclosure with reference to the drawings. Note that the disclosure is merely an example, and appropriate changes, which maintain the spirit thereof and are easily conceivable by those skilled in the relevant art, are naturally included in the scope of the present disclosure. Moreover, in some cases, in the present specification and the respective drawings, the same reference numerals are assigned to elements similar to those mentioned above regarding the already-discussed drawings, and a detailed description thereof is omitted as appropriate.

[0026] FIG. 1 illustrates a side plan view of a cooling system 100 incorporating a first cooling device 102, according to an example. As described, cooling system 100 may include one or more cooling devices 102 which include two or more elements described in reference to cooling system 100. Cooling system 100 and cooling device 102 may include a water header pipe 110 that may include a plurality of drain holes 112 arranged in a row 114 that is parallel with a central axis 116 of water header pipe 110. Row 114 may be orientated at an angle 118 that is measured between a radius 120 to central axis 116 of water header pipe 110 and a vertical plane 122 that passes through central axis 116. For convenience, the structures described in the drawing figures may be compared with a three-dimensional, orthogonal coordinate system with an X-axis 104, a Y-axis 106, and a Z-axis 108. As illustrated, vertical plane 122 may correspond to a Y-Z plane (106, 108). Water header pipe 110 may be configured to receive a supply of water 124 and discharge a first portion of water 128 through plurality of drain holes 112. Angle 118 may be between at least 10-degrees to not more than 40-degrees. For a thicker cooling pad (e.g., X-axis 104 dimension) and water header pipe centered, angle 118 may be greater than 10-40 degrees. For a thinner cooling pad, angle 118 may be less than 10-40 degrees.

[0027] Cooling device 102 may include an evaporative cooling medium 140 comprising a first media portion 142 and a second media portion 144. Suitable evaporative cooling media 140, such as the one or more evaporative cooling pads comprising first media portion 142 and second media portion 144, may be obtained from Munters Corporation of Amesbury, MA 01913, USA. Suitable evaporative cooling pads may include cellulose material, such as cellulose paper, cardboard, or other material that may derive from natural or manufactured fiber sources such as plants, trees, tree bark, and the like. First media portion 142 may have the shape of a filled, 3-dimensional “box” or rectangular prism held in a frame, with a first media first end 150 oriented as a top-most portion of first media portion 142 in FIG. 1, a first media second end 152 oriented as a bottom-most portion of first

media portion 142 in FIG. 1, and a first media body 154 disposed between first media first end 150 and first media second end 152. First media first end 150 may be configured to receive first portion of water 128. First media portion 142 may include a first plurality of channels 146 within first media portion 142 arranged in a substantially vertical manner, e.g., substantially along Y-axis 106, as illustrated. As used herein, the term substantially may be used as similar terms effectively or practically. Hence, the term substantially along X-axis 104 or along Y-axis 106 may include the sense that the relative alignment of these elements may be the same as these reference directions in an effective or practical sense, without precise measuring, to allow for small variations while performing the same operations, for example. First media body 154 may be configured to receive and conduct first portion of water 128 within first media body 154 as a second portion of water 156. First media second end 152 may be configured to discharge a third portion of water 158.

[0028] Similarly, second media portion 144 may have the shape of a filled, 3-dimensional “box” or rectangular prism held in a frame, with a second media first end 170, a second media second end 172, and a second media body 174 disposed between second media first end 170 and second media second end 172. Second media first end 170 may be configured to receive and conduct third portion of water 158 within second media body 174 as a fourth portion of water 176. Second media body 174 may include a second plurality of channels 148 within second media portion 144 arranged in a substantially horizontal manner, e.g., substantially along X-axis 104, as illustrated. In this manner, fourth portion of water 176 may be distributed horizontally (e.g., laterally) within second media body 174 beginning with an exterior portion of second media body 174, as will be described more fully below. Thus, second media body 174 may become saturated with water received from supply of water 124, through first media body 154, and discharged from first media body 154 to second media portion 144. Second media body 174 may receive and conduct fourth portion of water 176 so that second media body 174 may receive an inflow of air 180 at a first temperature 182 (e.g., ambient temperature 182) and discharge an outflow of air 184 at a second temperature 186 that is lower than the first temperature, where outflow of air 184 may be cooled by evaporative cooling of or by fourth portion of water 176. In this manner, inflow of air 180 may pass through second media body 174 which contains water and emerge as outflow of air 184 traveling laterally along X-axis 104, as illustrated. As used herein, the process of running water over or through an evaporative cooling pad and drawing air through the evaporative cooling pad may lower the ambient dry bulb temperature of the in-flowing air to provide adiabatic cooling of the out-flowing air from the pad, for example.

[0029] Some portion of inflow of air 180 may travel over first media first end 150 and may result in a cross-flow component of air flow across first media first end 150 in a direction from an outside surface of first media portion 142 to an inside surface of first media portion 142, as will be described more fully below. Second media body 174 may be configured to distribute received fourth portion of water 176 and discharge a fifth portion of water 178 which may collect in a reservoir 192 (e.g., a tray) as a collected portion of water 194. As described, third portion of water 158 may be introduced initially near an edge portion of second media

portion 144, fourth portion of water 176 may extend from the edge portion of second media portion 144 toward an opposite edge so that fifth portion of water 178 may be emitted from an entirety of second media second end 172. A pump 196 may be operatively coupled with collected portion of water 194 (e.g., at least partially submerged) and configured to receive and transport collected portion of water 194 and provide supply of water 124 through a conduit, pipe, or other structure back to water header pipe 110. Thus, collected portion of water 194 may provide a supply of water 124 that may flow through evaporative cooling medium 140 to cool inflow of air 180 by evaporative cooling and then be recycled. Additional water may be supplied to compensate for water lost due to evaporation.

[0030] According to an example, cooling system 100 and cooling device 102 may further include an elongated deflector shield 160 disposed to at least partially surround (e.g., placed at least partially around) an upper portion 162 of water header pipe 110. Elongated deflector shield 160 may be configured to at least one of protect upper portion 162 of water header pipe 110 from impact damage and contaminants, and an inside or underside portion of elongated deflector shield 160 may redirect water from water header pipe 110 in a direction toward first media first end 150. Elongated deflector shield 160 may also include a lower portion 164 with a lower edge portion 166 that is laterally displaced from a second vertical plane 126 that is coplanar (e.g., a plane parallel with) with an outer surface 145 of first media portion 142 and parallel with vertical plane 122. In this manner, some amount of first portion of water 128 that may contact an inside/underside surface of elongated deflector shield 160 or an inside surface of lower portion 164 may be redirected toward first media first end 150. First portion of water 128 may be directed toward first media first end 150 by reflection off the inside surface of lower portion 164 or may fall under the force of gravity in a vertical manner (e.g., opposite Y-axis 106 direction, as illustrated) onto first media first end 150.

[0031] As described above, the discharge point of water from water header pipe 110 from plurality of drain holes 112 at angle 118 may not be directly downward (e.g., opposite Y-axis 106 direction). Instead, first portion of water 128 may be discharged at angle 118 and may be directed at least partly towards an outside edge of first media portion 142, as will be described more fully below. Various benefits and technical effects may accrue from this arrangement, where first portion of water 128 may be discharged at an angle, instead of directly downward. Such benefits of displacing plurality of drain holes 112 from a directly downward row position (e.g., a negative Y-axis 106 direction) to an angularly displaced row position as described may include reducing the likelihood that dirt or other contaminants that may reside or collect in supply of water 124 or which may accumulate in a bottom portion of water header pipe 110 may block passage of water through plurality of drain holes 112. Some debris, minerals, or sediment may accumulate within water header pipe 110 and may be removed during periodic maintenance, but such accumulated sediment would not immediately block plurality of drain holes 112, for example. Further, displacing plurality of drain holes 112 from a directly downward row position may provide an exterior surface of water header pipe 110 that may be easier to clean, be less likely to corrode, and be more long-lasting. Additionally, first portion of water 128 may be more likely to

contact first media first end 150 if some amount of first portion of water 128 is displaced by a cross-flow component of air flowing across first media first end 150. Stated differently, a higher proportion of discharged first portion of water 128 may be more likely to land on and be captured by first media first end 150 resulting in more water captured within first media portion 142 and subsequently conducted to second media portion 144, and less water waste resulting in higher recycle efficiency.

[0032] Cooling system 100 may further include a gas cooler 130 disposed adjacent to the cooling device 102, where gas cooler 130 may be configured to receive hot gas 132 (e.g., a heated coolant or refrigerant material such as carbon dioxide) at a gas cooler inlet 134 and provide cooled gas 136 at a gas cooler outlet 138. Gas cooler 130 may include one or more refrigerant conducting pipes 131 (e.g., coils) which may be surrounded by heat radiating fins 133, for example. Typically, a gas cooler is configured to reduce the temperature of a circulating coolant gas without changing the state of the coolant from gas to liquid. However, this is not considered limiting.

[0033] Separately, gas cooler 130 may be configured to use air to cool a circulating gas, while cooling device 102 may be configured to reduce a surrounding air temperature. Together, cooling device 102 and gas cooler 130 may comprise a gas cooling unit 139 in cooling system 100, where cooling device 102 may reduce surrounding air temperature that is applied to gas cooler 130. In this manner, cooling device 102 may be considered as a “pre-cooler” for gas cooler 130 and may efficiently improve the operation of gas cooler 130. Second media body 174 may receive and conduct fourth portion of water 176 so that second media body 174 may receive inflow of air 180 at a first temperature 182 and discharge outflow of air 184 at a second temperature 186 that is lower than the first temperature, where outflow of air 184 may be cooled by evaporative cooling of fourth portion of water 176. As mentioned above, the process of running water over or through an evaporative cooling pad such as second media portion 144 and drawing air through the evaporative cooling pad may lower the ambient dry bulb temperature of the in-flowing air to provide adiabatic cooling of the out-flowing air from the pad. Outflow of air 184 may be applied to gas cooler 130 which may discharge a second outflow of air 188 at a second outflow temperature 190. Compared with a stand-alone gas cooler, the combination of cooling device 102 with gas cooler 130 provides a lower intermediate airflow temperature 186 compared with ambient temperature 182 without cooling device 102.

[0034] FIG. 2 illustrates a front plan view of water header pipe 110, according to an example. In addition to the description of FIG. 1, FIG. 2 illustrates how water header pipe 110 may include plurality of drain holes 112 arranged in row 114 that is parallel with central axis 116 (e.g., center of cylinder) of water header pipe 110. Further, water header pipe 110 may receive water from supply of water 124 into a center portion 202 of water header pipe 110, a first end portion 204 of water header pipe 110, or a second end portion 206 of water header pipe 110. As described, center portion 202 may be any position between first end portion 204 and second end portion 206. A coupling from supply of water 124 to water header pipe 110 may be made using a T-fitting, a sleeve fitting, or an angle fitting, for example. Alternatively, supply of water 124 may be coupled with a combination of components such as a pipe or other conduit

from another water header pipe (not shown in this view). At least one end portion (204, 206) of water header pipe 110 may be closed and/or capped to prevent outflow of water and to maintain pressure during expulsion of first portion of water 128 from water header pipe 110 from pump 196, for example. Pump 196 may operate according to a pump profile (e.g., water volume and water pressure) that is limited to at least one of: avoid overspray, avoid reflecting or bouncing off (e.g., ricochet) first media first end 150, to reduce water waste, and promote capture and conduction of water in the first media portion 142.

[0035] FIG. 3 illustrates an enlarged side plan view of a portion of the cooling system of FIG. 1, according to an example. In addition to the description of FIGS. 1-2, FIG. 3 illustrates how first portion of water 128 may be directed indirectly or directly to a first media portion exterior edge 302 (e.g., a top-front corner) of first media portion 142. In this manner, an exterior side portion 306 in the X-axis 104 direction of first media portion 142 may become saturated (e.g., filled with water from first portion of water 128) first so that an outside portion of first media portion 142 may have a relatively higher water content (e.g., or a higher water density gradient) compared with other portions of first media portion 142, for example. The substantially vertical orientation of first plurality of channels 146 within first media portion 142 may allow gravity to more quickly conduct first portion of water 128 through first media portion 142 to be discharged from first media second end 152 and thus enter second media first end 170 closer to a second media portion exterior surface 310 leading to second media body 174 being more reliably saturated with water on an outer surface exposed directly to air leading to more efficient cooling. First media portion 142 and/or second media portion 144 may include angled perforations that allow water to drain from the outer edges of the pads, effectively cleaning the external surface and removing accumulated dirt, for example. Further, angled perforations may limit some amount of water from entering the interior of the pads and which may limit water droplets being carried into gas cooler 130 and/or coils 131 via the airflow.

[0036] FIG. 4 illustrates a partial perspective view of a cooling system 400 incorporating the cooling device 102 of FIG. 1, according to an example. In addition to the descriptions of FIGS. 1-3, FIG. 4 illustrates cooling system 400 may include cooling system 102 with water header pipe 110 fed from supply of water 124 at a center portion 202 with a T-fitting, first media portion 142, a second media portion 144, and a gas cooler 130. As described above, water header pipe 110 may provide first portion of water 128 through first media portion 142 and third portion of water 158 to second media portion 144. In this example, cooling system 400 may include a fan portion 406 configured to expel air from cooling system 400 in a vertical (e.g., a Y-axis 106) direction, thereby pulling or drawing air through both a water-saturated second media portion 144 and gas cooler 130.

[0037] FIG. 5 illustrates a partial perspective view of an elongated deflector shield 160 partially surrounding an upper portion 162 of water header pipe 110 of the cooling system of FIG. 1, according to an example. In this example, water header pipe 110 may be fed by supply of water 124 at second end portion 206, for example. Elongated deflector shield 160 may be mounted (e.g., suspended) above water header pipe 110 by suitable mounting hardware, for example. In various examples, water header pipe 110 and/or

elongated deflector shield 160 may include various metal such as aluminum, steel, and stainless steel and/or plastic materials such as resin, a thermoplastic, or polyvinyl chloride (PVC).

[0038] FIG. 6 illustrates a side plan view of a cooling system 600 incorporating a second cooling device 602, according to an example. In addition to the description of FIGS. 1-5, FIG. 6 illustrates cooling system 600 may include one or more cooling devices 602 which include two or more elements described in reference to cooling system 600. Similar to cooling system 100 and cooling device 102, cooling system 600 and cooling device 602 may include a second water header pipe 610 that may include a second plurality of drain holes 612 arranged in a second row 614 that is parallel with a second central axis 616 of water header pipe 610. As described, many elements of second cooling device 602 may be similar to or identical with corresponding elements of first cooling system 100 and/or first cooling device 102, where at least some elements and features may be reflected about Y-axis 106, as illustrated. Second row 614 may be orientated at a second angle 618 that may be measured between a second radius 620 to second central axis 616 of second water header pipe 610 and a second vertical plane 622 that is parallel to first vertical plane 122 and that passes through second central axis 616. Second water header pipe 610 may be configured to receive supply of water 124 and discharge a sixth portion of water 628 through second plurality of drain holes 612. Second angle 618 may be between at least 10-degrees to not more than 40-degrees.

[0039] Cooling device 602 may include an evaporative cooling medium 640 comprising a third media portion 642 and a fourth media portion 644. Third media portion 642 may have the shape of a filled, 3-dimensional “box” or rectangular prism, with a third media first end 650 oriented as a top-most portion of third media portion 642 in FIG. 6, a third media second end 652 oriented as a bottom-most portion of third media portion 642 in FIG. 6, and a third media body 654 disposed between third media first end 650 and third media second end 652. Third media first end 650 may be configured to receive sixth portion of water 628. With brief reference to FIG. 3 and FIG. 6, sixth portion of water 628 may be directed to a third media portion exterior edge (e.g., reference 302). Third media body 642 may include a third plurality of channels 646 within third media portion 642 arranged in a substantially vertical manner, e.g., substantially along Y-axis 106, as illustrated. Third media body 654 may be configured to receive and conduct sixth portion of water 628 within third media body 654 as a seventh portion of water 656. Third media second end 652 may be configured to discharge an eighth portion of water 658. As illustrated, second angle 618 may be arranged on an opposite side of (e.g., symmetrically reflected) second vertical plane 622 which is parallel with first vertical plane 122 as compared with first angle 118. Thus, both first row 114 and second row 614 may be discharge first portion of water 128 and sixth portion of water 628, respectively, outward (e.g., toward an outer edge) of second media portion 144 and fourth media portion 644, respectively. With brief reference to FIG. 3 and FIG. 6, eighth portion of water 658 may be configured to initially saturate a fourth media portion exterior surface (e.g., reference 310).

[0040] Similarly, fourth media portion 644 may have the shape of a filled, 3-dimensional “box” or rectangular prism,

with a fourth media first end 670, a fourth media second end 672, and a fourth media body 674 disposed between fourth media first end 670 and fourth media second end 672. Fourth media first end 670 may be configured to receive and conduct eighth portion of water 658 within fourth media body 674 as a ninth portion of water 676. Fourth media body 674 may include a fourth plurality of channels 648 within fourth media portion 644 arranged in a substantially horizontal manner, e.g., substantially along X-axis 104, as illustrated. In this manner, ninth portion of water 676 may be distributed horizontally (e.g., laterally) within fourth media body 674 beginning with an exterior portion of fourth media body 674, as will be described more fully below. Thus, fourth media body 674 may become saturated with water received from supply of water 124, through third media body 654, and discharged from third media body 654 to fourth media portion 644. Fourth media body 674 may receive and conduct ninth portion of water 676 so that fourth media body 674 may receive a second inflow of air 680 at a fourth temperature 682 (e.g., ambient temperature 682) and discharge a third outflow of air 684 at a fifth temperature 686 that may be lower than fourth temperature 682, where third outflow air 684 may be cooled by evaporative cooling of ninth portion of water 676. In this manner, second inflow of air 680 may pass through fourth media body 674 which contains water and emerge as third outflow of air 684 traveling laterally along X-axis 104 (e.g., in an opposite direction), as illustrated. Some portion of second inflow of air 680 may travel over third media first end 650 and may result in a cross-flow component of air flow across third media first end 650 in a direction from an outside surface of third media portion 642 to an inside surface of third media portion 642. Fourth media body 674 may be configured to distribute received ninth portion of water 676 and discharge a tenth portion of water 678 which may collect in reservoir 192 as collected portion of water 194 along with discharged fifth portion of water 178.

[0041] As described, eighth portion of water 658 may be introduced initially near an edge portion of fourth media portion 644, ninth portion of water 676 may extend from the edge portion of fourth media portion 644 toward an opposite edge so that tenth portion of water 678 may be emitted from an entirety of fourth media second end 672. As mentioned above, pump 196 may receive and transport collected portion of water 194 to provide supply of water 124. In this manner, collected portion of water 194 may provide supply of water 124 that may flow through evaporative cooling medium 640 to cool second inflow of air 680 by evaporative cooling and then be recycled.

[0042] According to an example, cooling system 600 and second cooling device 602 may further include a second elongated deflector shield 660 disposed to at least partially surround (e.g., positioned at least partially around) a second upper portion 662 of second water header pipe 610. Second elongated deflector shield 660 may be configured to at least one of protect second upper portion 662 of second water header pipe 610 from impact damage and contaminants, and an interior or underside portion of second elongated deflector shield 660 may redirect discharged water from second water header pipe 610 in a direction toward third media first end 650. Second elongated deflector shield 660 may include a second lower portion 664 with a second lower edge portion 666 that is laterally displaced from a fourth vertical plane 626 that is parallel with a second outer surface 645 of third

media portion 642 and parallel with second vertical plane 622. In this manner, some amount of sixth portion of water 628 that may contact an inside or underside surface of second elongated deflector shield 660 or an inside or underside surface of second lower portion 664 may be redirected toward third media first end 650. Sixth portion of water 628 may be directed toward third media first end 650 by reflection off the inside or underside surface of second lower portion 664 or may fall under the force of gravity in a vertical manner (e.g., opposite Y-axis 106 direction) onto third media first end 650.

[0043] As described above, the discharge point of water from second water header pipe 610 from second plurality of drain holes 612 at second angle 618 may not be directly downward (e.g., opposite Y-axis 106 direction). Instead, sixth portion of water 628 may be discharged at second angle 618 and be directed at least partly towards an outside edge of third media portion 642. Various benefits and technical effects may accrue from this arrangement, where sixth portion of water 628 may be discharged at an angle, instead of directly downward. As described above, such benefits of displacing second plurality of drain holes 612 from a directly downward row position to an angularly displaced row position as described may include reducing the likelihood that dirt or other contaminants that may reside or collect in supply of water 124 which may accumulate in a bottom portion of second water header pipe 610 may block passage of water through second plurality of drain holes 612. Further, an exterior surface of water header pipe 110 may be easier to clean, be less likely to corrode, and be more long-lasting. Additionally, sixth portion of water 628 may be more likely to contact third media first end 650 if displaced by a cross-flow component of air flow across third media first end 650 resulting in higher recycle efficiency.

[0044] Cooling system 600 may further include a second gas cooler 630 disposed adjacent to second cooling device 602, where second gas cooler 630 may be configured to receive hot gas 632 at a second gas cooler inlet 634 and provide cooled gas 636 at a second gas cooler outlet 638. Second gas cooler 630 may include one or more refrigerant conducting pipes 631 surrounded by heat radiating fins 633, for example. Separately, second gas cooler 630 may be configured to use air to cool a circulating gas, while cooling system 600 and second cooling device 602 may be configured to reduce a surrounding air temperature. Together, second cooling device 602 and second gas cooler 630 may comprise a second gas cooling unit 139, where cooling device 602 may reduce surrounding air temperature that is applied to second gas cooler 630. In this manner, second cooling device 602 may be considered as a “pre-cooler” for second gas cooler 630 and may efficiently improve the operation of second gas cooler 630. As mentioned above, fourth media body 674 may receive and conduct ninth portion of water 676 so that fourth media body 674 may receive second inflow of air 680 at a fourth temperature 682 and discharge third outflow of air 684 at a fifth temperature 686 that is lower than fourth temperature 682, where third outflow of air 684 may be cooled by evaporative cooling of ninth portion of water 676. Third outflow of air 684 may be applied to second gas cooler 630 which may discharge a fourth outflow air 688 at a fifth outflow temperature 690. Compared with a stand-alone gas cooler, the combination of second cooling device 602 with second gas cooler 630 provides a lower initial airflow temperature 686 compared

with ambient temperature 682 without second cooling device 602. Under various conditions, first temperature 182, second temperature 186, third temperature 190, fourth temperature 682, fifth temperature 686, and sixth temperature 690 may be the same or different from each other.

[0045] FIG. 7 illustrates an end view of a cooling system 700 incorporating two cooling devices, according to an example. In addition to the descriptions of FIGS. 1-6, FIG. 7 illustrates cooling system 700 that may include first cooling device 102 having first water header pipe 110 disposed on a first side of cooling system 700 (e.g., a right-side, as illustrated) to provide water to first evaporative cooling media 140 adjacent to first gas cooler 130. Cooling system 700 may also include a second cooling device 602 that includes second water header pipe 610 disposed on a second side of cooling system 700 (e.g., a left-side as illustrated), opposite first cooling device 102, to provide water to second evaporative cooling media 640 adjacent to second gas cooler 630. In this manner, cooling system 700 may have two cooling devices (102, 602) and two gas coolers (130, 630) having similar or identical properties as these same elements described in reference to FIGS. 1-6, for example. In this example, first evaporative cooling media 140 and second evaporative cooling media 640 may both span the width of first gas cooler 130 and second gas cooler 630 on their respective sides of cooling system 700. First evaporative cooling media 140 may include multiple instances (e.g., panels) of first media portion 142 and multiple instances of second media portion 144 (shown in FIG. 1), for example. Similarly, second evaporative cooling media 640 may include multiple instances (e.g., panels) of third media portion 642 and multiple instances of fourth media portion 644 (shown in FIG. 6), for example.

[0046] First water header pipe 110 and second water header pipe 610 may be coupled together via a coupling pipe 711, where either first water header pipe 110 or second water header pipe 610 may receive water directly from supply of water 124, for example. Alternatively, both first water header pipe 110 and second water header pipe 610 may be sourced directly from supply of water 124. In this example, first inflow of air 180 from a first direction at a first temperature 182 (e.g., ambient temperature 182) may flow through at least a portion of first cooling device 102 and then through first gas cooler 130. Similarly, second inflow air 680 from a second direction, which may be substantially opposite the first direction of first inflow of air 180, may flow through at least a portion of second cooling system 602 and second gas cooler 630. Other arrangements of cooling system 600 are possible, where the cooling devices (102, 602) may be disposed on adjacent faces of cooling system 600, for example. Cooling system 700 may also include a fan 750 configured to expel exhaust air 760 in a vertical (e.g., a Y-axis 106) direction, for example. In this manner, first inflow of air 180 and second inflow of air 680 may be drawn from outside cooling system 700 into an inner cavity 706 and combined, then expelled as exhaust air 760. Cooling system 700 may include one fan 750 or a plurality of fans, as will be described below.

[0047] FIG. 8 illustrates a side view of a dual-fan cooling system 800 incorporating two cooling devices, according to an example. In addition to the descriptions of FIGS. 1-7, FIG. 8 illustrates a cooling system 800 that may include a first cooling device 102 having a first water header pipe 110 disposed on a first side of cooling system 800 (e.g., a

foreground as illustrated) to provide water to first evaporative cooling media 140 adjacent to a first gas cooler 130. Cooling system 800 may also include a second cooling device 602, similar to first cooling device 102, having second water header pipe 610 disposed on a second side of cooling system 800 (e.g., a background as illustrated), opposite first cooling device 102, to provide water to second evaporative cooling media 640 adjacent to second gas cooler 630. In this example, first evaporative cooling media 140 illustrated in the foreground of FIG. 8 and second evaporative cooling media 640 illustrated in the background of FIG. 8 may both span the width of first gas cooler 130 and second gas cooler 630 on their respective sides of cooling system 800. First evaporative cooling media 140 may include multiple instances (e.g., panels) of first media portion 142 and multiple instances of second media portion 144 (shown in FIG. 1), for example. Similarly, second evaporative cooling media 640 may include multiple instances (e.g., panels) of third media portion 642 and multiple instances of fourth media portion 644 (shown in FIG. 6), for example. First water header pipe 110 and second water header pipe 610 may be coupled together via a coupling pipe 711, where either first water header pipe 110 or second water header pipe 610 may receive water from supply of water 124 to provide water to evaporative cooling media 140 for both first cooling device 102 and second cooling device 602, for example. Alternatively, both first water header pipe 110 and second water header pipe 610 may be sourced from supply of water 124 that is also coupled to coupling pipe 711 to provide water to evaporative cooling media 140 for both first cooling device 102 and second cooling device 602. Cooling system 800 may also include first fan 750 and a second fan 751 configured to expel first exhaust air 760 and second exhaust air 761 in a vertical (e.g., a Y-axis 106) direction, for example. In this manner, first inflow of air 180 and second inflow of air 680 may be drawn from outside cooling system 800 into an inner cavity 706 and combined, then ejected as first exhaust air 760 and second exhaust air 761. Inner cavity 706 may be continuous or divided vertically between first fan 750 and second fan 751. An advantage of inner cavity 706 being continuous is that air may be drawn through an entirety of first evaporative cooling media 140 and second evaporative cooling media 640 by first fan 750 if second fan 751 is inoperable, or vice versa.

[0048] FIG. 9 illustrates a side view of a 7-fan cooling system 900 incorporating two cooling devices, according to an example. In addition to the descriptions of FIGS. 1-8, FIG. 9 illustrates a cooling system 900 that may include a first cooling device 102 having a first water header pipe 110 disposed on a first side of cooling system 900 (e.g., a foreground as illustrated) to provide water to first evaporative cooling media 140 adjacent to a first gas cooler 130. Cooling system 900 may also include a second cooling device 602, similar to first cooling device 102, having a second water header pipe 610 disposed on a second side of cooling system 900 (e.g., a background as illustrated), opposite first cooling device 102, to provide water to second evaporative cooling media 640 adjacent to second gas cooler 630. In this example, first evaporative cooling media 140 illustrated in the foreground of FIG. 9 and second evaporative cooling media 640 illustrated in the background of FIG. 9 both span the width of first gas cooler 130 and second gas cooler 630 on their respective sides of cooling system 900. In this example, first evaporative cooling media 140 illus-

trated in the foreground of FIG. 9 and second evaporative cooling media 640 illustrated in the background of FIG. 9 may both span the width of first gas cooler 130 and second gas cooler 630 on their respective sides of cooling system 900. First evaporative cooling media 140 may include multiple instances (e.g., panels) of first media portion 142 and multiple instances of second media portion 144 (shown in FIG. 1), for example. Similarly, second evaporative cooling media 640 may include multiple instances (e.g., panels) of third media portion 642 and multiple instances of fourth media portion 644 (shown in FIG. 6), for example. First water header pipe 110 and second water header pipe 610 may be coupled together via a coupling pipe 711, where either first water header pipe 110 or second water header pipe 610 may receive water from supply of water 124 to provide water to evaporative cooling media 140 for both first cooling device 102 and second cooling device 602, for example. Alternatively, both first water header pipe 110 and second water header pipe 610 may be sourced from supply of water 124 that is also coupled to coupling pipe 711 to provide water to evaporative cooling media 140 for both first cooling device 102 and second cooling device 602. Cooling system 900 may also include a plurality of fans (e.g., first fan 750, second fan 751, . . . , seventh fan 756) configured to expel exhaust air (e.g., first exhaust air 760, second exhaust air 761, . . . , seventh exhaust air 766) in a vertical (e.g., a Y-axis 106) direction, for example. In this manner, first inflow air and second inflow air may be drawn from outside cooling system 900 into an inner cavity 706 and combined, then expelled as exhaust air. Inner cavity 706 may be continuous or divided vertically between first fan 750 through seventh fan 756.

[0049] FIG. 10 illustrates a method 1000 of operating a cooling system, according to an example. In addition to the descriptions of FIGS. 1-9, FIG. 10 illustrates a method 1000 (e.g., a process) that may begin with a step of discharging 1006 first portion of water 128 from water header pipe 110 with plurality of drain holes 112 arranged in row 114 that may be parallel with central axis 116 of water header pipe 110. Row 114 may be oriented at angle 118 measured between radius 120 to central axis 116 of water header pipe 110 and vertical plane 122 that passes through central axis 116. Method 1000 may continue with a step of receiving 1010 first portion of water 128 at first media first end 150 of first media portion 142. Method 1000 may continue with a step of conducting 1014 first portion of water 128 within first media body 154 of first media portion 142 as second portion of water 156. First media body may include first plurality of channels 146 arranged in a substantially vertical manner 106 (e.g., a y-axis 106 direction, as illustrated). Method 1000 may continue with a step of discharging 1018 second portion of water 156 as third portion of water 158 from first media second end 152 of first media portion 142. Method 1000 may continue with a step of receiving 1022 third portion of water 158 at second media first end 170 of second media portion 144. Method 1000 may continue with a step of conducting 1026 third portion of water 158 within second media body 174 as fourth portion of water 176. Second media body 174 may include second plurality of channels 148 arranged in a substantially horizontal manner 104 (e.g., an X-axis 104 direction, as illustrated).

[0050] Method 1000 may continue with a step of receiving 1030 first inflow of air 180 at a first temperature 182 into second media body 174. Method 1000 may continue with a

step of cooling 1034 the first inflow of air by evaporative cooling of (e.g., provided by) fourth portion of water 176 in second media body 174 to provide first outflow of air 184 at second temperature 186 that is lower than first temperature 182, for example. Method 1000 may continue with a step of discharging 1038 first outflow of air 184. Angle 118 may be between at least 10-degrees to not more than 40-degrees. First portion of water 128 may be directed to a first media portion exterior edge. Fourth portion of water 176 may be configured to initially saturate a second media portion exterior surface. Method 1000 may continue with a step of discharging 1042 fourth portion of water 176 as fifth portion of water 178 from second media second end 172 of second media portion 144. Method 1000 may continue with a step of collecting 1046 fifth portion of water 178 as collected fifth portion of water 194. Method 1000 may continue with a step of pumping 1052 at least a portion of collected fifth portion of water 194 to provide supply of water 124 to water header pipe 110. Method 1000 may continue by returning to step 1006, for example. Simultaneously, a parallel method may continue with second water header pipe 610 and the structure described in reference to FIGS. 1-9, in various examples.

[0051] Modifications, additions, or omissions may be made to processes of method 1000 depicted in FIG. 10 associated with FIGS. 1-9, as described. Such processes may include more, fewer, or other operations. For example, various operations may be performed in parallel or in any suitable order, where such parallel operation or reordering is not prohibited by the above descriptions. Further, the cooling systems and methods described herein may also be performed as a method of exchanging heat. Finally, the described methods may be performed in conjunction with a computer readable medium including instructions that when executed cause a controller or computer to operate a cooling system, as described.

[0052] While several examples have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

[0053] In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

[0054] To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. 112(f) as it

exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

What is claimed is:

1. A cooling device, comprising:
 - a water header pipe having a plurality of drain holes arranged in a row that is parallel with a central axis of the water header pipe, the row being oriented at an angle measured between a radius to the central axis of the water header pipe and a vertical plane passing through the central axis, the water header pipe being configured to receive a supply of water and discharge a first portion of water through the plurality of drain holes; and
 - an evaporative cooling medium comprising a first media portion and a second media portion,
 - the first media portion having a first media first end, a first media second end, and a first media body, the first media first end being configured to receive the first portion of water, the first media body being configured to conduct the received first portion of water within the first media body as a second portion of water, the first media second end being configured to discharge a third portion of water; and
 - the second media portion having a second media first end, a second media second end, and a second media body, the second media first end being configured to receive the third portion of water, the second media body being configured to conduct the received third portion of water within the second media body as a fourth portion of water.

wherein the second media body is configured to receive a first inflow of air at a first temperature and discharge a first outflow of air at a second temperature that is lower than the first temperature, the first outflow of air being cooled by evaporative cooling of the fourth portion of water.
2. The cooling device of claim 1, wherein the angle is between at least 10-degrees to not more than 40-degrees.
3. The cooling device of claim 1, wherein the first portion of water is directed to a first media portion exterior edge, and wherein the second portion of water is configured to initially saturate a second media portion exterior surface.
4. The cooling device of claim 1,
 - wherein the first media body includes a first plurality of channels arranged in a substantially vertical manner, and
 - wherein the second media body includes a second plurality of channels arranged in a substantially horizontal manner, orthogonal to the first plurality of channels.
5. The cooling device of claim 1, wherein at least one of the first media portion and the second media portion include cellulose.
6. The cooling device of claim 1, further comprising:
 - an elongated deflector shield disposed to at least partially surround an upper portion of the water header pipe, the elongated deflector shield being configured to at least one of protect the upper portion of the water header pipe and redirect water from the water header pipe in a direction toward the first media first end.
7. The cooling device of claim 6, wherein the elongated deflector shield includes a lower portion with a lower edge portion that is laterally displaced from a second vertical plane that is coplanar with an outer surface of the first media portion.

8. The cooling device of claim 1, wherein the second media second end is configured to discharge a fifth portion of water, the device further comprising:

- a reservoir configured to collect the fifth portion of water as a collected fifth portion of water; and
- a pump configured to transport the collected fifth portion of water and provide the supply of water to the water header pipe.

9. The cooling device of claim 8, wherein the pump has a pump profile that is limited to at least one of avoid overspray, avoid bouncing off first media first end, and promote capture and conduction of water in the first media portion.

10. The cooling device of claim 8, wherein the supply of water is provided to one of a center portion of the water header pipe and an end portion of the water header pipe, wherein at least one end portion of the water header pipe is closed.

11. A cooling system, comprising the cooling device of claim 1, the system further comprising:

- a gas cooler disposed adjacent to the evaporative cooling medium and configured to receive the first outflow of air and provide a second outflow of air.

12. A cooling system comprising the cooling device of claim 1, wherein the water header pipe is a first water header pipe and the evaporative cooling medium is a first evaporative cooling medium, the system further comprising:

- a second water header pipe having a second plurality of drain holes arranged in a second row that is parallel with a second central axis of the second water header pipe, the second row being oriented at a second angle measured between a second radius to the second central axis of the second water header pipe and a second vertical plane passing through the second central axis, the second water header pipe being configured to receive the supply of water and discharge a sixth portion of water through the second plurality of drain holes; and

a second evaporative cooling medium comprising a third media portion and a fourth media portion,

- the third media portion having a third media first end, a third media second end, and a third media body, the third media first end being configured to receive the sixth portion of water, the third media body being configured to conduct the received sixth portion of water within the third media body as a seventh portion of water, the third media second end being configured to discharge an eighth portion of water; and

the fourth media portion having a fourth media first end, a fourth media second end, and a fourth media body, the fourth media first end being configured to receive the eighth portion of water, the fourth media body being configured to conduct the received eighth portion of water within the fourth media body as a ninth portion of water,

wherein the fourth media body is configured to receive a second inflow of air at a fourth temperature and discharge a third outflow of air at a fifth temperature that is lower than the fourth temperature, the third outflow of air being cooled by evaporative cooling of the ninth portion of water.

13. The cooling system of claim 12, wherein the second angle is between at least 10-degrees to not more than

40-degrees, wherein the sixth portion of water is directed to a third media portion exterior edge, wherein the eighth portion of water is configured to initially saturate a fourth media portion exterior surface, wherein at least one of the third media portion and the fourth media portion include cellulose.

14. The cooling system of claim **12**, further comprising:
a second elongated deflector shield disposed to at least partially surround a second upper portion of the second water header pipe, the second elongated deflector shield being configured to at least one of protect the second upper portion of the second water header pipe and redirect water from the second water header pipe in a direction toward the third media first end, wherein the second elongated deflector shield includes a second lower portion with a second lower edge portion that is laterally displaced from a fourth vertical plane that is coplanar with a second outer surface of the third media portion.

15. The cooling system of claim **12**, further comprising:
a first gas cooler disposed adjacent to the first evaporative cooling medium and configured to receive the first outflow of air and provide a second outflow of air.
a second gas cooler disposed adjacent to the second evaporative cooling medium and configured to receive the third outflow of air and provide a fourth outflow of air.

16. The cooling system of claim **15**, further comprising:
at least one fan unit configured to receive the second outflow of air and the fourth outflow of air and expel a fifth outflow of air.

17. A cooling method, comprising:
discharging a first portion of water from a water header pipe with a plurality of drain holes arranged in a row that is parallel with a central axis of the water header pipe, the row oriented at an angle measured between a radius to the central axis of the water header pipe and a vertical plane that passes through the central axis;

receiving the first portion of water at a first media first end of a first media portion;

conducting the first portion of water within a first media body of the first media portion as a second portion of water, the first media body includes a first plurality of channels arranged in a substantially vertical manner;

discharging the second portion of water as a third portion of water from a first media second end of the first media portion;

receiving the third portion of water at a second media first end of a second media portion;

conducting the third portion of water within a second media body as a fourth portion of water, the second media body includes a second plurality of channels arranged in a substantially horizontal manner;

receiving a first inflow of air at a first temperature into the second media body;

cooling the first inflow of air by evaporative cooling of the fourth portion of water in the second media body to provide a first outflow of air at a second temperature that is lower than the first temperature; and

discharging the first outflow of air.

18. The cooling method of claim **17**, wherein the angle is between at least 10-degrees to not more than 40-degrees.

19. The cooling method of claim **17**, wherein the first portion of water is directed to a first media portion exterior edge, and wherein the fourth portion of water is configured to initially saturate a second media portion exterior surface.

20. The cooling method of claim **17**, further comprising:
discharging the fourth portion of water as a fifth portion of water from a second media second end of the second media portion;

collecting the fifth portion of water as a collected fifth portion of water; and

pumping at least a portion of the collected fifth portion of water to provide a supply of water to the water header pipe.

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